ALGEBRA 1 · UNIT 10: FACTORING AND ZERO PRODUCT PROPERTY

Packaging Engineer

Factor area and volume expressions to design boxes and labels.

CP1

2–3 class periods

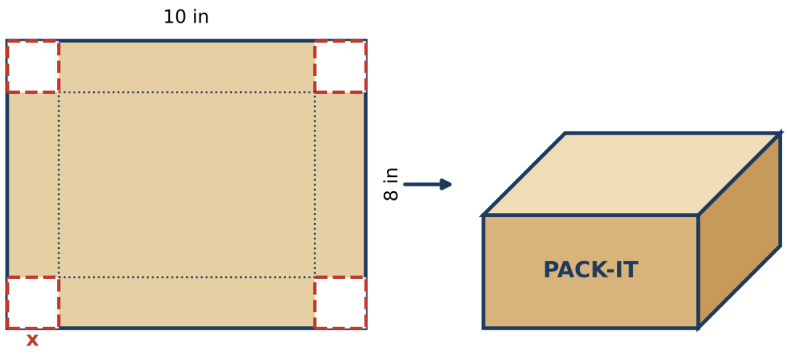
Name: _____ Date: _____ Period: _____

The Scenario

Pack-It Packaging designs boxes, labels, and posters for local businesses. Customers usually tell the company the AREA or VOLUME they need, and the engineers must work backward to find the dimensions.

Working backward from area is factoring! When a customer gives an exact area, you will set up a quadratic equation and solve it with the Zero Product Property. Then you will design a box from a flat sheet of cardboard.

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Learning Goals

- I can factor polynomials using GCF, trinomials, grouping, and special patterns.
- I can solve quadratic equations by factoring in context.
- I can explain which solutions are reasonable.

Project Timeline

When	What you will do
Day 1	Area & Volume to Dimensions
Day 2	Exact Size Orders
Day 3	Design a Box
Final	Box Design Spec Sheet + reflection

Part 1: Area & Volume to Dimensions

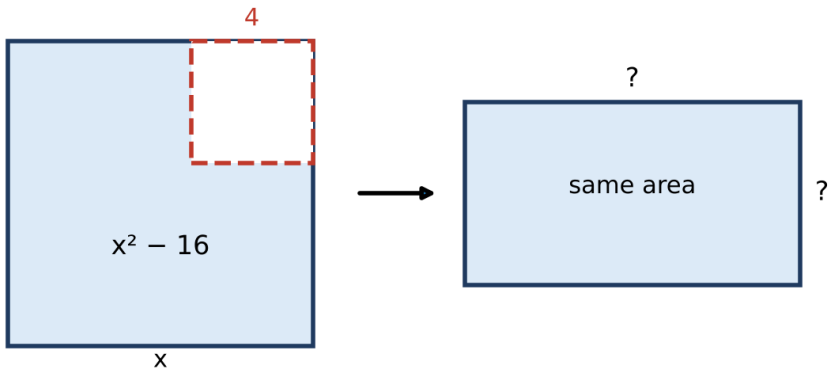
Day 1 · Factor completely to find dimensions.

1. A box has volume $2x^3 + 10x^2 + 12x$. Factor completely to find three dimensions.

2. Labels have areas $2x^2 + 7x + 3$ and $3x^2 - 10x - 8$. Factor each.

3. A shipping crate has volume $x^3 + 3x^2 + 4x + 12$. Factor by grouping.

4. Explain, using the cut-square picture, why $x^2 - 16 = (x + 4)(x - 4)$.



Part 2: Exact Size Orders

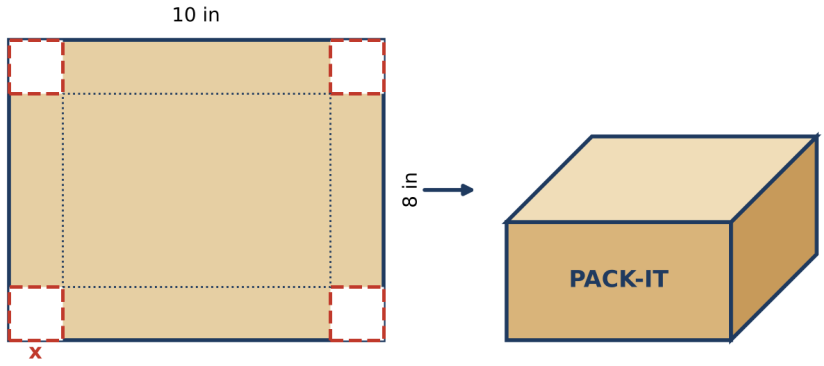
Day 2 · Write an equation, solve by factoring, and reject any solution that doesn't make sense.

5. A label's length is 3 inches more than twice its width. Its area is 44 square inches. Find its dimensions.

6. A 20 in by 30 in poster gets a frame of width x all the way around. The framed poster has an area of 704 square inches. Find x.

7. Cut squares of side x from a 10 in by 8 in sheet. The base must be 48 square inches. Find x and the volume.

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8. Solve $3x^2 - 10x = 8$ and decide which solution could be a length.

Part 3: Design a Box

Day 3 · Design an open box from a rectangular sheet of your choice (whole-number dimensions).

9. Choose your sheet size and a target base area that gives a whole-number cut size. Write and solve the equation.

10. Find the box's dimensions and volume. Explain why one solution must be rejected.

11. Write a spec sheet entry giving the base area in factored form and the volume as a factored polynomial in x.

Final Product: Box Design Spec Sheet

Your final product must include:

- Factored dimensions for every label and box
- Solutions to all exact-size orders with rejected solutions explained
- Your box design with equation, dimensions, and volume
- A spec sheet with factored expressions

Reflection

R1. How do you decide which factoring method to try first?

R2. Why does a frame problem always give one negative solution?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Box design meets the customer's area or volume exactly, and impossible solutions are rejected with a reason. Goes beyond the minimum with a creative, realistic choice.	Box design meets the customer's area or volume exactly, and impossible solutions are rejected with a reason.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Spec sheet shows the factored dimensions, the equation solved with the Zero Product Property, and a labeled net or drawing. Polished, organized, and visually engaging.	Spec sheet shows the factored dimensions, the equation solved with the Zero Product Property, and a labeled net or drawing. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.